සියලු ම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடையது/All Rights Reserved]

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2022(2023)
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ව්‍යාපාර සංඛ්‍යානය	I
வணிகப் புள்ளிவிவரவியல்	I
Business Statistics	I

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පැය දෙකයි
இரண்டு மணித்தியாலம்
Two hours

Instructions:

- *Answer **all** questions.*
- *Write your **Index Number** in the space provided in the answer sheet.*
- *Statistical tables will be provided. Use of calculator is **not allowed**.*
- *Carefully follow the Instructions given on the back of the answer sheet.*
- *In each of the questions **1** to **50**, pick one of the alternatives from (1), (2), (3), (4), (5) **which is correct or most appropriate** and **mark your response on the answer sheet with a cross** (×) on the number of the correct option in accordance with the instructions given at the back of the answer sheet.*

1. Which of the following statements is true?

(1) Statistical decisions can be taken based on both aggregate and individual observations.
(2) The subject, statistics can be considered as the study of only the average.
(3) Any phenomenon can be proved by the statistics.
(4) Statistical data may be misused due to lack of care and unawareness of users.
(5) Results of a certain statistical study can be valid for any situation considered.

2. Consider the following statements.

A - The main objective of a pre-test is to test the clarity, accuracy, consistency and completeness of the questionnaire.
B - True zero can exist in both interval scale and ratio scale data.
C - Coding ranks given to successive points of an ordinal scale should not necessarily be according to a certain scale.

The true statement/s from the above is/are,

(1) only A. (2) only A and B. (3) only A and C.
(4) only B and C. (5) all A, B and C.

3. Which of the following statements is false?

(1) Any characteristic which can be directly expressed in numerical values is called as quantitative data.
(2) The accuracy and reliability of primary data are relatively very high even if it takes more time and cost.
(3) The secondary data which is used for a study should not necessarily be relevant to the purpose of the study.
(4) Personal interview method may be more efficient if the interviewer is an experienced and well trained person.
(5) Focus group discussion is more appropriate for collecting qualitative data related to attitudes, beliefs and experiences of people.

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4. Consider the following statements.
   A - There is no difference between class limits and class boundaries in a group frequency distribution with overlapping classes.
   B - Data can be summarized by a stem and leaf plot and a box-and-whiskers plot.
   C - Grouping error occurs when the group frequency distribution is constructed by assuming all the observations are coincided on the corresponding class mark.

   The true statement/s from the above is/are,
   (1) only A. (2) only B. (3) only A and B.
   (4) only A and C. (5) all A, B and C.

5. Which of the following statements is **false**?
   (1) Lorenz curve can be used to compare a specific situation in income inequality with the common situation.
   (2) A Z-chart can be used to observe short term variations, totals up to a certain time point and the trend of a considered variable.
   (3) Data relevant to a certain variable presented in several pie charts can be presented in a single percentage component bar chart.
   (4) Percentiles of a data distribution can be calculated by using relative cumulative frequency distribution.
   (5) A multiple bar chart can be used to present changes of a certain variable with several related components.

6. The mean of the weekly sales of rice at a particular wholesale rice shop is 2500kg. The mean of the weekly sales of white and red rice are 2700kg and 1700kg respectively. If the rice shop sells only the above two types of rice, then the percentages of weekly sales of white and red rice are respectively,
   (1) 20 and 80 (2) 20.88 and 79.22 (3) 49.22 and 50.88
   (4) 50.88 and 49.22 (5) 80 and 20

7. Consider the following data set.
   127, 162, 138, 192, 144, 177, 154, 141, 232, 144, 171, 152, 146, 132
   The mean, median and the mode of the above data set are respectively,
   (1) 144, 149 and 158 (2) 149, 144 and 158 (3) 149, 158 and 144
   (4) 158, 144 and 149 (5) 158, 149 and 144

8. Consider following statements about the measures of central tendency.
   A - The geometric mean is a commonly used measurement which is used to calculate the average of the rate of change or ratios.
   B - All the three measurements mean, median and mode can be affected by extreme values of the data distributions.
   C - Mean, median and mode of any data distribution can be obtained graphically as well as mathematically.

   The true statement/s from the above is/are,
   (1) only A. (2) only A and B. (3) only A and C.
   (4) only B and C. (5) all A, B and C.

9. Which of the following statements is **false**?
   (1) Standard deviation can be affected by extreme values of the data distribution.
   (2) Coefficient of variation is considered as a measure of dispersion which does not have a unit impact.
   (3) If Karl Pearson's first coefficient of skewness (S_{k1}) lies between -0.5 and $+0.5$, the data distribution can be considered as a moderately skewed distribution.
   (4) The quartile deviation is an appropriate measure for comparing dispersion of different data distributions with extreme values.
   (5) If the coefficient of kurtosis is a greater value for a certain data distribution, the coefficient of skewness also may be a greater value.

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10. Consider the following stem-and-leaf diagram.

3	4								
4	3	4	7						
5	2	2	4	5	7				
6	2	2	3	4	4	7	9		
7	2	3	3	4	5	5	6	7	7
8	0	1	1	4					

Semi inter-quartile range and Bowley's coefficient of skewness of the above data are respectively,

(1) 10.5 and –4 (2) 10.5 and –0.19 (3) 10.5 and 0.19

(4) 21 and –0.19 (5) 21 and 0.19

11. If mean, coefficient of variation and Karl Pearson's coefficient of skewness for a skewed data distribution are 300, 12 and 0.4 respectively, the mode of the data distribution is,

(1) 285.6 (2) 295.2 (3) 296.4 (4) 304.8 (5) 314.4

12. Which of the following statements is true about correlation and regression?

(1) Scatter diagram can be used to examine only the linear relationship between two variables.

(2) Coefficient of determination is equal to the squared value of the Spearman's rank correlation coefficient.

(3) Karl Pearson's product moment correlation coefficient can be used to measure the relationship between two quantitative variables.

(4) Freehand method can be used to make accurate and reliable predictions in regression analysis even if there is no mathematical base.

(5) The estimated regression coefficient of a simple linear regression model gives the number of units changed in the dependent variable relative to one unit changed in the independent variable.

13. Consider the following statements about regression.

A - Fitting a straight line by minimizing the sum of squares of vertical deviations of each points on straight line from the observed values is called the least squares method

B - If $y=\beta_0+\beta_1 x+u$ is the regression model for a certain population, u denotes the error term that describes the effects on y of all factors except x.

C - A relationship that completely determines the values of dependent variable by the given values of independent variable is called as stochastic relationship.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.

(4) only A and C. (5) all A, B and C.

14. The estimated regression equation for a given data set is $\hat{y}=15.8-0.128x$. The total variation and error variation of the model are 25.20 and 4.78 respectively.
Consider the following statements related to the above information.

A - There is a strong negative linear relationship between X and Y even if the estimated regression coefficient is very small.

B - The coefficient of correlation (r) between X and Y is approximately -0.9.

C - The goodness of fit of the estimated regression equation is very poor since the regression coefficient value is small.

The true statement/s from the above is/are,

(1) only A. (2) only A and B. (3) only A and C.

(4) only B and C. (5) all A, B and C.

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15. Consider the following statements about the approaches to probability

A - The classical approach can be applied for the possible outcomes of the random experiments which are unable to do repeatedly under the same conditions.

B - Two persons can never get the same result under the subjective approach to probability.

C - If two events X and Y are collectively exhaustive in a sample space S, under the mathematical approach to probability it is required that the $P(X)+P(Y)=1$.

The false statement/s from the above is/are,

(1) only A. (2) only A and B. (3) only A and C.
(4) only B and C. (5) all A, B and C.

16. If A and B are any two events in the sample space S such that $P(A)=\frac{1}{3}$, $P(B)=\frac{1}{4}$ and $P(A\cup B)'=\frac{7}{12}$ then the value of $P(B|A')$ is

(1) $\frac{1}{18}$ (2) $\frac{1}{16}$ (3) $\frac{1}{8}$ (4) $\frac{1}{4}$ (5) $\frac{2}{3}$

17. A and B are any two events such that $P(A)=p$, $P(B)=\frac{p}{2}$ and $P(A'\cap B)+P(A\cap B')=\frac{3}{5}p$.

If the two events A and B are independent then the value of $P(B)$ is,

(1) $\frac{1}{5}$ (2) $\frac{9}{20}$ (3) $\frac{3}{5}$ (4) $\frac{4}{5}$ (5) $\frac{9}{10}$

18. If the probability function for a sample space of a certain random experiment is given as $P(x)=\frac{1}{42}(5x+3)$, the possible values for the sample space are,

(1) $-2,-1,0$ and 1 (2) $-1,0,1$ and 2 (3) $0,1,2$ and 3
(4) $1,2,3$ and 4 (5) $2,3,4$ and 5

19. The random variable X has the following probability distribution.

X	-3	-2	-1	2	3
$P(X=x)$	0.2	0.15	p	0.3	0.2

Which of the following statements for the random variable X is true?

(1) The expected value of the random variable X is negative.
(2) Standard deviation of X can be negative since the negative values of X.
(3) The probability of getting a negative value for X is twice as much as probability of getting a positive value for X.
(4) The smallest value of X for which $P(X\leq x)>0.5$ is 2.
(5) The random variable X is satisfied the condition $E[X^2]<[E[X]]^2$

20. If the random variable X has a binomial distribution with $E[X+1]=7$ and $V(X+2)=3.6$, then $P(X\geq 2)$ is,

(1) 0.0032 (2) 0.0047 (3) 0.0052 (4) 0.9729 (5) 0.9948

21. A manager of a particular company receives a large stock of goods. He decided to accept the stock if the number of defective goods is no more than 2 by taking a random sample of 200 goods. If he knows that upon his past experiences 2% of goods will be defective, then the probability of rejecting the stock is,

(1) 0.0916 (2) 0.1465 (3) 0.2381 (4) 0.7619 (5) 0.9084

[see page five

Weights of packets of sugar has a normal distribution with a mean of 500 grams. If 10% of sugar packets are under the weight of 487.2 grams, then the percentage of sugar packets which weight more than 502 grams is,

(1) 1.28 (2) 7.93 (3) 42.07 (4) 57.93 (5) 92.07

23. Consider the following statements about the simple random sampling.

A - For most situations, the estimator for a sample with replacement is less efficient than the estimator for a sample without replacement.

B - Selecting a sample with keeping an equal probability to each unit of the population to be included, is called simple random sampling.

C - If the sampling fraction is small, the true variance can be reduced by ignoring the finite population correction factor.

The true statement/s from the above is/are,

(1) only A. (2) only A and B. (3) only A and C.
(4) only B and C. (5) all A, B and C.

24. Which of the following statements is true about the sampling?

(1) Stratified random sampling can be used even when a sampling frame is not available.
(2) Simple random sampling is more suitable for a population with a linear trend.
(3) If the variation within clusters is small, cluster sampling is more appropriate.
(4) If the variation among strata is small, stratified sampling is more suitable.
(5) If the sampling fraction is small, then the finite population correction factor will be close to one.

25. In simple random sampling without replacement a sample is drawn from a population with the known variance σ^2. If the sample mean and variance can be considered to be $\overline{X}$ and S^2 respectively, then the standard error of the sample mean with sample size n is,

(1) $\frac{\sigma}{\sqrt{n}}$ (2) $\sqrt{\frac{\sigma^2}{n}\left(\frac{N-n}{N}\right)}$ (3) $\sqrt{\frac{S^2}{n}\left(\frac{N-n}{N}\right)}$

(4) $\sqrt{\frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)}$ (5) $\sqrt{\frac{S^2}{n}\left(\frac{N-n}{N-1}\right)}$

26. Which of the following statements is false?

(1) Since the mean ($\overline{X}$), median (M_d) and mode (M_o) of a normal distribution are equal the sample median and mode are unbiased estimators for the mean μ of the Normal distribution.
(2) If equal size samples have been drawn, the estimator with the least variance among the unbiased estimators is called the most efficient estimator.
(3) When increasing the sample size n, if the estimator is centered about the unknown parameter, it is called a consistent estimator.
(4) When comparing with the sample mean ($\overline{X}$), the sample median (X_m) is a more efficient estimator for unknown parameter μ.
(5) If all the sample data have been used to compute the estimator which estimates the unknown parameter μ, it is called a sufficient estimator.

27. $X_1, X_2, X_3, \ldots, X_n$ is a random sample from a population with mean μ and variance σ^2. If $T = \frac{\sum_{i=1}^{n} x_i}{n-1}$ is a estimator for μ, the biasedness of the estimator T is,

(1) $\frac{\mu}{n-1}$ (2) $\frac{(1-n)\mu}{n-1}$ (3) μ (4) $\frac{n\mu}{n-1}$ (5) $\frac{(2n+1)\mu}{n-1}$

28. If the 95% confidence interval estimated for the mean of a normal population with mean μ and $\sigma^2 = 256$ is 152.08, 159.92, the size of the sample taken to construct the interval is,

(1) 8 (2) 9 (3) 32 (4) 64 (5) 128

29. 10, 8, 12, 6, 14 are sample values from a normal population. If the sample variance is 10, the margin of error of the interval estimator for the population mean μ at 95% confidence level is,

(1) $1.96\sqrt{2}$ (2) $2.02\sqrt{2}$ (3) $2.78\sqrt{2}$ (4) $2.13\frac{10}{\sqrt{5}}$ (5) $2.78\frac{10}{\sqrt{5}}$

30. Consider the following statements about interval estimation.

A - Confidence level and the width of the confidence interval are positively related.
B - Sample size and the width of the confidence interval are inversely related.
C - To estimate a highly reliable and highly precise confidence interval, a larger sample and a greater level of confidence should be used.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only B and C. (5) all A, B and C.

31. Consider the following statements.

A - When increasing the sample size, the mean of the sampling distribution of a sample proportion is closed to the population proportion.
B - When the number of sample elements with a certain attribute is expressed as a ratio to the total number of population elements it is called as population proportion.
C - The probability distribution of the sample proportions of all possible samples with equal size from a certain population is called as sampling distribution of the sample proportion.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only A and C. (5) all A, B and C.

32. The number of motor cars coming to two service centers *A* and *B* per hour follow two independent poisson distributions with means $\lambda_A = 2.15$ and $\lambda_B = 1.75$ respectively. If 100 time intervals of one hour are considered, the sampling distribution of the difference between two sample means $(\overline{X}_A - \overline{X}_B)$ is approximately,

(1) $N\left(0.4, \left(\frac{2.15^2}{100} - \frac{1.75^2}{100}\right)\right)$ (2) $N\left(0.4, \left(\frac{2.15}{100} + \frac{1.75}{100}\right)\right)$

(3) $N\left(0.4, \sqrt{\left(\frac{2.15}{100} - \frac{1.75}{100}\right)}\right)$ (4) $N\left(0.4, \left(\frac{2.15^2}{100} + \frac{1.75^2}{100}\right)\right)$

(5) $N\left(0.4, \sqrt{\left(\frac{2.15}{100} + \frac{1.75}{100}\right)}\right)$

33. The following information were drawn from two independent samples with equal population variances.

Sample size (n)	Mean ($\overline{X}$)	Standard Deviation (S)
12	27.2	4
10	32.1	6

When constructing a 95% confidence interval for the difference of population means, the most suitable estimate for the variance is,

(1) 4.9 (2) 5 (3) 22.7 (4) 23.8 (5) 25

34. Consider the following statements about hypotheses testing.

A - The power of a hypothesis testing is the probability of rejecting a false null hypothesis.
B - The power of a hypothesis test can not be increased by decreasing standard error of the test statistics.
C - A person claims that 70% of the people in a certain city support his policies. To test his claim, two tailed hypothesis test is more appropriate.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only C.
(4) only A and B. (5) all A, B and C.

35. Two students A and B conducted two hypotheses testing for a certain problem. The student A conducted one tail hypothesis test and claimed that the result is significant at 3% level. The student B conducted a two tail test and claimed that the result is not significant at the same level. If both students obtained the same Z test statistic for these tests then the value of the test statistic is,
(1) -2.29 (2) 1.68 (3) 1.71 (4) 2.13 (5) 2.21

36. To test the hypothesis that 'a certain coin is fair', 64 tosses of the coin were made. If 24 heads were obtained out of total tosses, then the p-value of the test is,
(1) 0.0228 (2) 0.0456 (3) 0.4544 (4) 0.9544 (5) 0.9772

37. Which of the following statement is true about the p-value of a hypothesis test?
(1) The p-value depends on the significant level of a hypothesis test.
(2) If the p-value of the test is smaller than to significant level, then the alternative hypothesis should be rejected.
(3) The distribution of the test statistic need to be known to calculate the p-value of a test.
(4) The maximum probability of type-I error is known as p-value of a test.
(5) p-value can be increased by increasing the critical value of a hypothesis test.

38. A manufacturer claimed that the mean weight of his tea packets is 50 grams. To test this, a random sample of size 81 packets is taken and measured the weights of packets (X). X has a normal distribution with mean μ and σ=10. If the critical region for testing $H_0: \mu = 50$ against $H_1: \mu = 45$ is given by $\overline{X} < 48$, then the probability of accepting his claim when it is not true is,
(1) 0.0035 (2) 0.4965 (3) 0.5035 (4) 0.9641 (5) 0.9965

39. Three of the machine operators in a factory were randomly selected and recorded the number of units of a product produced by them in five successive months to test whether the mean productivity of them are equal. The calculated sum of the squares of totals (SST) and sum of the squares of errors (SSE) were as 352 and 156 respectively. The $F-$ statistic calculated to test the above hypothesis is,
(1) 2.659 (2) 4.776 (3) 5.025 (4) 7.538 (5) 13.538

40. Consider the following statements about time series analysis.
A - Original time series data may sometimes be adjusted to the changes in calender, price and population.
B - Multiplicative model is not suitable for the situation where each component of the time series depends on each other.
C - Method of semi-averages can be used to estimate trend accurately even though extreme values are presented in the time series.
The true statement/s from the above is/are,
(1) only A. (2) only A and B. (3) only A and C.
(4) only B and C. (5) all A, B and C.

41. Which of the following statements is **false**?
(1) The method of ratio to percentages cannot be recommended to estimate seasonal indices if the long term trend is available in a time series.
(2) Trend equation estimated by the method of least squares can be used to predict trend values of any period of time.
(3) When converting an annual trend equation into a monthly trend equation of a time series, both gradient and intercept of the equation may be changed.
(4) The cyclical movements of a time series can be identified from the boom existed over an one year period on the long term trend.
(5) Least squares trend equation of a time series may be obtained by minimizing the total sum of squares $\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$ of the time series.

42. In the process of estimating seasonal component, if the original time series value (Y) for each month is divided by corresponding centered moving average, the remaining components are,
(1) T and C. (2) S and I. (3) T, C and I. (4) T, S and I. (5) C, S and I.

43. The trend equation estimated for monthly production (in '000) of a garment factory with 15th July of 2018 as the origin is given by $\hat{Y}=132+8t$. If the seasonal index for the October of 2021 is 120 and assume a multiplicative model, the estimated production for the October of the year 2021 is approximately,

(1) 358 000 (2) 378 000 (3) 422 400 (4) 429 600 (5) 432 000

44. The weight per packet of coffee powder is monitored by a quality control process. Control limits of the process are set at $UCL_{\overline{X}}=20.12$ grams and $LCL_{\overline{X}}=19.90$ grams. If samples of size five are used for the inspection process, the process mean and standard deviation for the manufacturing operation are respectively,

(1) 20.01 and 0.069 (2) 20.01 and 0.082 (3) 20.01 and 0.191
(4) 40.02 and 0.069 (5) 40.02 and 0.082

45. Consider the following statements about statistical quality control.

A - The range chart examines the variations of ranges of the relevant samples are how much deviated from the mean range value.
B - Control charts are used to examine whether the raw materials and the finished goods are in accordance with the pre-determined standards.
C - The number of defective units in each sample variated about the mean number of defective units is depicted in the *C*-chart.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only A and C. (5) all A, B and C.

46. 12 samples with size five were taken from the manufacturing process. If the central line $CL_{\overline{X}}$ of the control chart and the sum of the sample ranges ($\sum_{i=1}^{k} R_i$) were 71.38 and 720 respectively, the lower and upper control limits for the $\overline{X}$-chart are respectively,

(1) 27.64 and 115.12 (2) 36.76 and 106.00 (3) 36.76 and 115.12
(4) 55.42 and 87.34 (5) 55.42 and 106.00

47. For an acceptance sampling plan with n=50 and c=2, the probability of rejecting a lot with 5% defectives is,

(1) 0.0154 (2) 0.4562 (3) 0.5438 (4) 0.7127 (5) 0.9846

48. Consider the following statements about index numbers.

A - Index numbers are used in studying the changes of a series over a period of time in prices, quantities or volumes.
B - In a situation of the construction of index numbers, weights refers to the absolute importance of commodities considered.
C - Fisher's ideal price index satisfies both time reversal and circular tests but does not satisfy the factor reversal test.

The true statement/s from the above is/are,

(1) only A. (2) only A and B. (3) only A and C.
(4) only B and C. (5) all A, B and C.

49. The earnings of a non-skilled worker in a factory was Rs. 24 000 in the year 2019 and was increased to Rs. 43 000 in the year 2021. The cost of living indices in 2019 and 2021 were 125 and 325 respectively. To maintain the same standard of living in 2021, the additional allowance he should have received is Rs.,

(1) 9 231 (2) 19 400 (3) 62 400 (4) 68 800 (5) 87 800

50. For the year 2018, the price relative indices of two commodities *X* and *Y* were 125 and 140 respectively. If the weighted price index for the year was 130, the percentage of the total weights given to commodity *Y* is,

(1) 33.00 (2) 33.33 (3) 33.67 (4) 66.33 (5) 66.67

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සියලු ම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடையது/All Rights Reserved]

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2022(2023)
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2022 (2023)
General Certificate of Education (Adv. Level) Examination, 2022 (2023)

ව්‍යාපාර සංඛ්‍යානය II வணிகப் புள்ளிவிவரவியல் II **Business Statistics II**	**31 E II**

පැය තුනයි மூன்று மணித்தியாலம் ***Three hours***	අමතර කියවීම් කාලය - මිනිත්තු 10 යි மேலதிக வாசிப்பு நேரம் - 10 நிமிடங்கள் **Additional Reading Time - 10 minutes**

Use additional reading time to go through the question paper, select the questions you will answer and decide which of them you will prioritise.

Instructions:

* *Answer five questions selecting at least two questions from each part.*
* *Statistical tables and graph papers will be provided. Use of calculators is* ***not allowed.***

Part I

1. (a) Describe the following methods of data collection giving **two** advantages and two disadvantages for each.

(i) Direct observation

(ii) Personal interview (06 *marks*)

(b) State the scale of measurement relevant to each of the following situation.

(i) Time taken to complete a mathematics paper in an examination.

(ii) Selecting a colour for motor cars.

(iii) Number of absences of a worker per year.

(iv) Rating community of a school as principal, sectional heads, teachers, prefects and students. (02 *marks*)

(c) (i) Explain **two** reasons for using different diagrammatic presentation methods to present data. (02 *marks*)

(ii) The frequency distribution given below is about the time length of phone calls made by 175 people during the weekend.

Length of a phone call (in minutes)	1-7	8-14	15-21	22-28	29-35	36-42	43-49	50-56
Frequency	45	32	34	22	16	12	9	5

Draw the histogram and frequency polygon on the same diagram and hence obtain the mode of the distribution. (04 *marks*)

(d) The following table provides the Gross Domestic Production (GDP) annual growth rates of a certain country from 2010 to 2019.

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
GDP growth rate	8.0	8.4	9.1	3.4	5.0	5.0	4.5	3.6	3.3	2.3

(i) Draw a line graph for the above data. (02 *marks*)

(ii) Comment on the behaviour of the GDP annual growth rate observing the line graph. (03 *marks*)

(iii) State **two** limitations of use of a line graph. (01 *mark*)

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2. (a) (i) Explain the meaning of mean, median and mode as measures of central tendency. (03 *marks*)

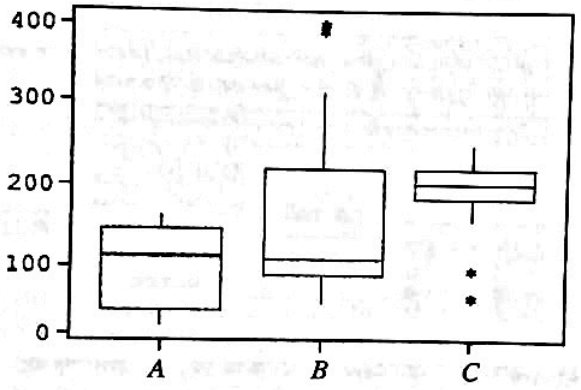
(ii) Describe what is the most appropriate measure of central tendency for the data distributions in each of nominal, ordinal, interval and ratio scales of measurement. (04 *marks*)

(b) Selected several summary measures related to a certain data distribution are given below.

$Q_1 = 40$ $Q_3 = 90$ $P_{10} = 30$ $P_{90} = 110$ $\overline{X} = 45$ $M_d = 70$ $S = 35$

(i) Calculate an appropriate measurement for each of the coefficient of skewness and coefficient of kurtosis. (02 *marks*)

(ii) The data distributions related to annual profits (Rs. Billion) of three companies *A*, *B* and *C* for a certain time period are presented in following Box-and-whiskers plot.



State **three** comments on characteristics of annual profits distributions of three companies by comparing the above Box-and-whiskers plots. (03 *mark*)

(c) Number of overtime hours of workers in a small scale factory for a month are given in following table.

No. of overtime hours	No. of workers (*f*)
30 - 40	4
40 - 50	12
50 - 60	24
60 - 70	34
70 - 80	15
80 - 90	6
90 - 100	5
	100

(i) Calculate mean, median, mode, variance and Karl Pearson's first coefficient of skewness of the distribution. (07 *marks*)

(ii) Comment on the shape of the data distribution based on above results. (01 *mark*)

3. (a) A company buys four products A, B, C and D. The prices and ordered quantities of products for years 2020 and 2021 are given in following table.

Product	2020		2021	
	Price	Quantity	Price	Quantity
A	19	28	20	34
B	23	55	25	51
C	10	20	11	24
D	17	63	17	84

(i) Calculate Laspeyre's and Paasche's price indices for the year 2021, taking 2020 as the base year. (04 *marks*)

(ii) What is the most appropriate price index from the above two price indices? Explain. (01 *mark*)

(iii) Calculate the Fisher's price index using the results of the part (i) above. (01 *mark*)

(b) (i) What is a cost of living index?

List **two** uses of a consumer price index. (02 *marks*)

(ii) List **four** factors which should be considered in constructing a consumer price index. (02 *marks*)

(c) (i) What is meant by a time series?

Describe **two** uses of time series analysis in business field. (02 *marks*)

(ii) State **two** advantages and **two** disadvantages of moving average method as a method of estimating trend. (02 *marks*)

(d) The trend equation estimated by the method of least squares for the data of annual sales (in millions) of a soft drink company is given as $\hat{Y} = 656 + 32.96t$ (Origin is 2017).

(i) Convert the annual trend equation into the quarterly trend equation. (01 *mark*)

(ii) If the seasonal indices of third and fourth quarters of the year 2020 are 95 and 125 respectively, estimate sales for these two quarters. (04 *marks*)

(iii) Comment on estimated sales. (01 *mark*)

4. (a) Explain **two** uses of correlation and regression analysis in business field. (01 *mark*)

(b) A particular Light Rail Transit (LRT) project wanted to determine whether there is any relationship between the lifetime of a light train and the annual maintenance cost. The following table shows the lifetime and their maintenance costs of randomly selected six light trains.

Lifetime of light train in years (X)	1	2	3	4	5	6
Annual maintenance cost in Rs. million (Y)	0.2	1.0	1.4	1.4	2.0	2.4

(i) Fit the regression line of Y on X by the method of least squares. (04 *marks*)

(ii) Comment on $\hat{\beta}_0$ and $\hat{\beta}_1$. (01 *mark*)

(iii) What is the importance of coefficient of determination for a regression model? (01 *mark*)

(c) Eight candidates in an interview were ranked by three interviewers in the following order.

Interviewer	Rank							
A	2	4	5	3	1	7	8	6
B	2	4	6	3	1	8	7	5
C	4	3	7	2	1	5	8	6

If the Spearman's rank correlation coefficient between interviewer *A* and *B* was 0.95, find which pair of interviewers has the closer relationship, by calculating Spearman's rank correlation coefficient between other interviewers. (03 *marks*)

(d) (i) Explain the importance of statistical quality control in producing high-quality products. (01 *mark*)

(ii) Suppose that a buyer receives large stock of items. The acceptance sampling plan is to inspect a random sample of 100 items and c=2. Calculate the producer's risk and the consumer's risk if the stock has 1% and 5% defective items. (04 *marks*)

(iii) State the difference between np-chart and U-chart.

Number of defects observed in 10 motor cars manufactured by a vehicle manufacturing factory are as follows.

Motor car No.	1	2	3	4	5	6	7	8	9	10
Number of defects	4	6	6	4	12	10	8	10	18	12

Construct an appropriate control chart for the above data and state whether the motor car manufacturing process is in under control. (05 *marks*)

Part II

5. (a) Describe the relative frequency approach of probability. State two limitations of this approach. (03 *marks*)

(b) There are four women executive officers and five men executive officers in a certain department. Calculate the probability that at least two women executive officers must be selected for a committee which consists of five members. (02 *marks*)

(c) If two independent events A and B such that $P(B|A') = \frac{3}{5}$ and $P(A) = \frac{13}{20}$. Find,

(i) $P(B)$

(ii) $P(A' \cup B')$

(iii) Are two events A and B collectively exhaustive? (03 *marks*)

(d) The probability that a person will get a job as a management trainee of a multinational company is $\frac{2}{5}$ and the probability that he will not get a job as a government officer is $\frac{4}{7}$. If the probability of getting at least one job is $\frac{2}{3}$. What is the probability that he will get the job only as a management trainee? (03 *marks*)

(e) Bayes' theorem is an extension of the conditional probability. Explain. (02 *marks*)

A recent study has revealed that only 55% of people in a certain country who do not wear face masks suffer particular respiratory diseases, while only 20% of people who wear face mask suffer from the same respiratory disease. If 35% of people wear face masks, calculate the probability that a randomly selected person,

(i) wears face masks and suffers from the respiratory disease. (02 *marks*)

(ii) suffers from the respiratory disease. (02 *marks*)

(iii) does not wear face masks, given that the person suffers from the respiratory disease. (03 *marks*)

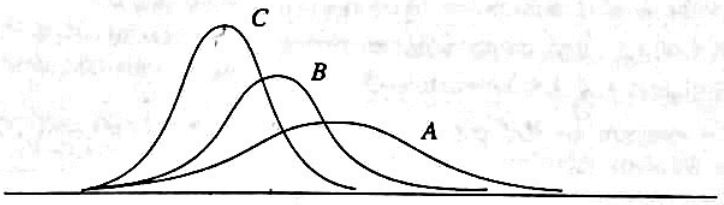
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6. (a) Giving reasons, state each of the following statement/s is/are true or false.
   (i) The binomial distribution is not that much needed since its values can always be approximated by another distribution.
   (ii) If the expected number of arrival of people in a particular office is calculated as seven per hour, then it can not be definitely said that seven people will arrive within the next hour.
   (iii) The mean always lies in between the mode and the median for a normal distribution.
   (iv) The corresponding value of Z for a point of X lying in a normal distribution is shown by the area between X and the mean of the distribution. (04 *marks*)

   (b) (i) Define the binomial distribution stating the conditions which should be satisfied for it. (02 *marks*)
   (ii) The committee of a certain company contains 10 members. Find the probability that more females than males will be members of the committee under the condition that at least one male member is in the committee when a female has 40% chance to be selected to the committee. (03 *marks*)

   (c) (i) State the conditions under which the normal distribution may be used as an approximation to poisson distribution. (01 *mark*)
   (ii) A certain manufacturing company faces power breakdowns which follows a poisson distribution with average of nine (9) per week. Assuming these breakdowns occur randomly, calculate the probability that there will be 40 breakdowns in any given month. (03 *marks*)

   (d) A large number of certain plants are planted in a plant nursery. The heights of the plants are recorded after two months, five months and seven months. If heights of these plants have a normal distribution and those data are shown by the three curves below,



   (i) Which curve represents the height distribution of plants after two months? (01 *mark*)
   (ii) What can be said about the mean height of the plants and the standard deviation with time progression? (02 *marks*)

   (e) The time taken for grade seven students to solve a small puzzle follows a normal distribution with a mean four minutes and standard deviation of 30 seconds. 10% of the students who spend least time on solving the puzzle are awarded grade '*A*' while the next 10% are awarded grade '*B*' and remaining students are awarded grade '*C*'. What is the maximum time boundary for grade '*A*' to be awarded? (04 *marks*)

7. (a) Differentiate between following pair of terms giving an example for each.
   (i) Sampling frame and sampling unit
   (ii) Sampling errors and non-sampling errors (06 *marks*)

   (b) (i) What is stratified random sampling? (01 *mark*)
   (ii) Explain how quota sampling relates to stratified random sampling and state **two** advantages and **two** disadvantages of quota sampling. (03 *marks*)

(c) For the purpose of selecting a better sampling plan to estimate the population mean $\bar{Y}$, assume that all the population observations $N=6$ are given as follows.

$y_1= 8,\ y_2= 12,\ y_3= 64,\ y_4= 43,\ y_5= 100,\ y_6= 85$

Following two sampling plans A and B with $n = 3$ are provided.

Plan A	(8, 12, 100), (8, 12, 85), (12, 64, 43) (64, 43, 100), (43, 100, 85), (100, 85, 8)

Plan B	(8, 12, 85), (8, 64, 100), (8, 43, 100), (8, 43, 85) (12, 64, 100), (12, 64, 85), (12, 43, 100), (64, 43, 85)

(i) Calculate the sample mean $\bar{y}$ for the samples of above each sampling plan. (04 *marks*)

(ii) Calculate the expected value of $\bar{y}$ for each sampling plan. (01 *mark*)

(iii) Which sampling plan is more suitable? Explain. (01 *mark*)

(iv) If the calculated population variance S^2 is approximately 1192, calculate the variance of $\bar{y}$ only using the formula. (01 *mark*)

(d) State the central limit theorem.

A fresh milk packing machine is automated so that the amount of fresh milk packed is a random variable with mean μ=150 ml and variance σ^2=7.7 ^{2} ml. If a random sample of n=121 milk packets packed by the machine is selected, what is the approximate probability that the sample mean lies between 148.6 ml and 151.4 ml? (03 *marks*)

4. (a) Explain the difference between each of the following pair of terms.

(i) Type-I error and type-II error

(ii) Null hypothesis and alternative hypothesis

(iii) Simple hypothesis and composite hypothesis

(iv) Right-tailed test and left-tailed test (04 *marks*)

(b) From random samples of 100 people in each city A and B, 60 and 50 people are found to be literated respectively.

(i) Construct a 95% confidence interval for the difference between true population proportions of literate people in cities A and B. (02 *marks*)

(ii) Test the hypothesis that 'the proportions of literate people in two cities A and B are equal' by using the confidence intervals in part (i). (02 *marks*)

(iii) What is the p-value of the above test? (03 *marks*)

(c) Explain what is meant by contingency coefficient C. (02 *marks*)

(d) Five coins were tossed 100 times and the following results were obtained.

No. of heads	0	1	2	3	4	5
Observed frequency	10	15	25	25	15	10

(i) Fit a binomial distribution for these data. (03 *marks*)

(ii) Write down hypotheses for testing the goodness of fit. (01 *mark*)

(iii) Test the goodness of fit at a 5% level of significance and state your conclusions. (03 *marks*)

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